

Date: August 15, 2025

Algorithms -1-

University of Aleppo

Student Name:

Model: A

Duration: 90 Minutes

Student Number:

Select the correct answer for each question:

1. Calculate the derivative of $f(x) = x^2$ then find the limit to infinity of the function $\lim_{x \rightarrow \infty} f(x)$
A. 14 B. 15 C. 16 D. 17 E. 17 [1 points]
2. Please calculate $\int f(x)dx$
A. 13 B. 14 C. 15 D. 16 [2 points]
3. Test
A. 13 B. 14 C. 15 D. 16 E. 17 [3 points]
4. Hello!
A. 1 B. 2 C. 3 D. 4 [2 points]
5. Calculate $\sum i$
A. $n * (n + 1)/3$ B. $n * (n - 1)/2$ [1 points]
6. Find the area of a rectangle of side $a = 6$
A. 34 B. 36 C. 40 D. 31 [1 points]
7. For any positive integer x what is the value of $\int \int \int x^5$?
A. 30 B. 40 C. 50 D. 60 [1 points]
8. Solve for x in the equation $2x + 6 = 0$
A. -1 B. -2 C. -3 D. 3 [1 points]
9. $x = 3 + 6$ what is the value of x ?
A. 7 B. 9 C. 8 D. 10 [1 points]
10. What is the derivative of x^2
A. x B. $2 * x$ [5 points]